Idea Proposal: Fatura Dedektifi (Bill Detective)

Finding which appliance uses the electricity from a single household meter

Mert Güngör, Group 14, Samsung Innovation Campus, Artificial Intelligence Course

1. Project Idea

An electricity bill shows one total. It does not show which appliance used how much, so a household cannot tell where to save. Energy costs are a real burden in Türkiye. In 2025, 19.5% of the population could not afford to keep their home adequately warm [1]. Heating is mostly gas, but electricity is the bill a household can cut appliance by appliance. Putting a separate meter on every appliance is expensive and impractical.

I want to build a model that takes only the total power signal of a home and estimates how much the fridge, the washing machine, the dishwasher, the kettle and the microwave used. This task is known as non-intrusive load monitoring (NILM). The result will be a web demo that turns one day of meter data into a breakdown per appliance in kWh and in TL. I will measure success per appliance with the mean absolute error in watts (MAE) and the error in total energy, on homes the model never saw in training.

2. Relevance to the Sustainable Development Goals

The project serves SDG 7, Affordable and Clean Energy. It relates most to target 7.3 (improve energy efficiency), because people can only reduce the consumption they can see, and more loosely to target 7.1 (affordable energy services), by helping households lower their bills. It also supports SDG 12.8 (information and awareness for sustainable lifestyles) and SDG 13, since lower electricity use means lower emissions.

3. Literature Examples

Kelly and Knottenbelt (2015) [2] applied three deep neural networks to this task for five appliances in UK homes. The networks achieved better average F1 scores than the classical methods (combinatorial optimisation and factorial hidden Markov models) and also worked on a house not seen in training.

Zhang et al. (2018) [3] proposed sequence-to-point learning. A convolutional network reads a window of the total signal and predicts the power of one appliance at the midpoint of the window. They report improvements of 84% and 92% on two error measures.

Both papers evaluate models on research benchmarks. Neither turns the result into a tool or a cost breakdown for a household. My project tests on several unseen REFIT homes at 1-minute resolution and shows the result in kWh and TL.

4. Data

I will use REFIT Electrical Load Measurements (Cleaned) from the University of Strathclyde, free under CC BY 4.0 [4]. It covers 20 UK homes between 2013 and 2015. Each home has the total power and 9 appliance channels in watts, recorded about every 8 seconds. The format is CSV, one file per home. The archive is 514 MB and holds about 6.9 GB and 119.5 million rows.

I downloaded the data and checked it. A fridge or freezer is monitored in 20 homes, a washing machine in 19, a microwave in 16, a dishwasher in 15 and a kettle in 15. I analysed two homes in detail. In House 2 the nine monitored appliances explain 32% of the total energy, and the dishwasher alone 13%. About 25% of the minutes in House 2 have no data. For preprocessing I will resample to 1 minute, remove windows with gaps, map the columns of each home to common appliance names and scale the values. I will split by home (training, validation and test homes) to avoid leakage. One limit is that the homes are British and I have no Turkish data to test on. Appliances and habits differ, so the TL figures in the demo will be an illustration.

5. Approach: Deep Learning

I choose deep learning with a machine learning baseline. The input is a raw time series in which appliances overlap and draw similar power. In House 2 the heaters of the washing machine and the dishwasher both draw about 2.1 to 2.2 kW. A model has to learn the shape of each appliance's cycle, which suits a 1D convolutional network (sequence-to-point). I will train one small network per appliance on free Colab.

I already computed two baselines by training on House 2 and testing on House 5 at 1-minute resolution. Predicting the average power of each appliance gives an MAE of 44.9 W for the kettle and 107.1 W for the dishwasher. Gradient boosting on window statistics lowers these to 17.6 W and 67.6 W. It does not help for the washing machine (44.3 to 43.6 W) or the microwave (30.2 to 30.0 W). Window statistics lose the order of the phases in a cycle (heating, washing, spinning), which a convolutional network keeps, so I expect it to help most there. My target is to beat gradient boosting on the same test homes, in both MAE and daily energy error, for at least four of the five appliances.

References

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[2] Kelly, J., & Knottenbelt, W. (2015). Neural NILM: Deep Neural Networks Applied to Energy Disaggregation. ACM BuildSys '15. https://doi.org/10.1145/2821650.2821672

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